

Reducing Missed College Buses Through Timely Bus Arrival Notifications

A smart notification system to keep students informed and on time.

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PROGRAMME

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INSTITUTION

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The Problem

CURRENT SITUATION

- | Students depend on college buses for daily transportation
- | Buses are delayed due to traffic, road diversions, or weather
- | Students only know the scheduled time, not the actual arrival time
- | Students either miss the bus or wait for a long time

CONSEQUENCES

- Late arrival to college
- Increased transportation expenses
- Stress for students and parents
- Frequent complaints to the transport office

Who Is Affected?

STUDENTS

Want accurate bus arrival information and need to avoid missing the bus

PARENTS

Need assurance about their child's travel safety and punctuality

BUS DRIVERS

Want students to be ready at bus stops — reducing delays and waiting time

TRANSPORT OFFICE

Wants fewer complaints and easier communication with students

COLLEGE MANAGEMENT

Decision Maker

Needs an affordable, easy-to-maintain, and reliable solution — no expensive infrastructure

How It Works

"A simple system that informs students when their college bus is about to reach their stop so they can be ready on time."



SMART BUS ALERT SYSTEM FOR COLLEGE STUDENTS

Real-time Bus Updates • Timely Alerts • On-time Travel

⚠ THE PROBLEM

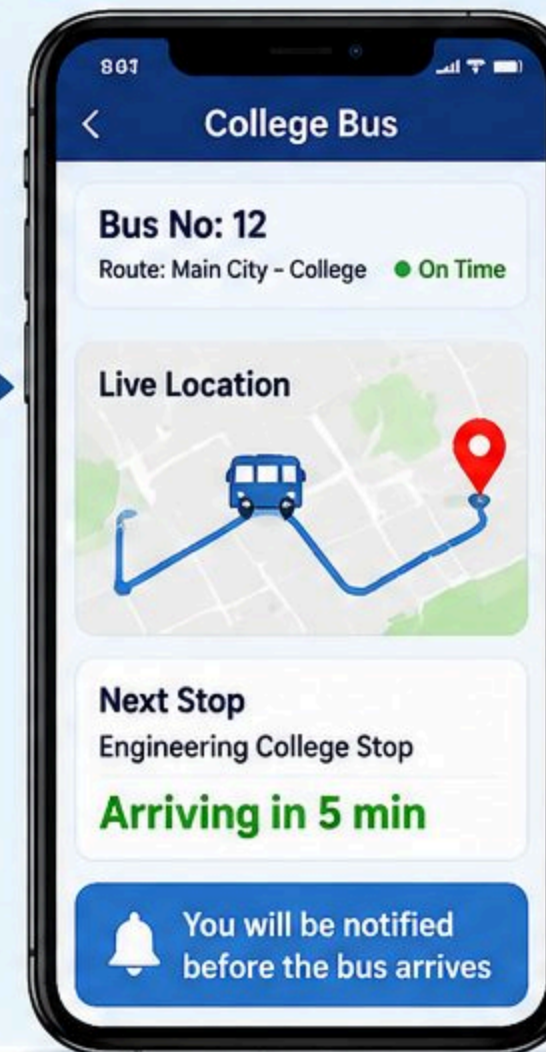


- ✗ Students miss the bus due to no real-time updates
- ✗ Long waiting time at bus stops
- ✗ Late arrival to college
- ✗ Frequent complaints to transport office
- ✗ Stress for students and parents

✅ OUR SOLUTION



Driver starts the bus and shares location in real-time



Students receive alert before the bus reaches their stop



Students reach the bus stop on time

HOW IT WORKS



1. Bus Starts



2. Location Updated



3. System Processes



4. Students Get



5. Student Reaches



6. On-time Pickup

KEY BENEFITS

- ✓ Fewer missed buses
- ✓ Better punctuality
- ✓ Reduced waiting time
- ✓ Less complaints
- ✓ Happy students & parents



Affordable & Easy to Adopt

AFFORDABLE

- | Uses drivers' existing smartphones
- | Requires only internet connectivity
- | Low maintenance cost
- | No expensive hardware installation

EASY TO ADOPT

- | Easy for students to use
- | Minimal training required
- | Reduces manual communication
- | Can be implemented in existing college transport services

Benefits

Fewer missed buses

Students arrive at stops before the bus

Improved punctuality

Students reach college on time consistently

Reduced complaints

Less pressure on the transport office

Better communication

Real-time bridge between students and transport office

Parent confidence

Families stay informed about travel status

Efficient transport management

Saves time, reduces overall inefficiency

Limitations & Future Improvements

CURRENT LIMITATIONS

- | Cannot prevent traffic congestion
- | Cannot avoid vehicle breakdowns
- | Depends on internet connectivity
- | Arrival time may vary due to unexpected conditions

FUTURE IMPROVEMENTS

- | Emergency delay notifications
- | Attendance integration
- | Route optimization
- | Parent notification feature
- | Transport analytics dashboard

Conclusion

- | Addresses a real campus transportation problem
- | Improves communication between students and the transport office
- | Affordable and easy to implement
- | Enhances punctuality and student convenience
- | Practical solution suitable for everyday college use



Thank You

We welcome your questions and feedback.